禁区里的生机

Life in the Forbidden Zone



禁区里动物变多，就能说明辐射无害吗?

英文原文

Chernobyl is synonymous with disaster, yet it holds a startling amount of wildlife. Early on 26 April 1986, one of its reactors exploded in Soviet Ukraine, dispersing radioactive fallout across Europe. On 27 April, nearby Pripyat was emptied of about 45,000 inhabitants within hours. A zone some 30 kilometres in radius became off-limits, and Belarus fenced off adjoining terrain.

In October 2015, a study in Current Biology combined aerial surveys with footprint counts on the Belarusian side. Elk, roe deer, red deer, and wild boar were comparable in density to those in four reserves free of contamination, and wolves were over seven times as numerous. Earlier helicopter counts showed elk, roe deer, and boar surging in the post-disaster decade.

Later fieldwork bolstered these findings. In April 2016, camera traps snapped wolves, boar, and raccoon dogs near the most hazardous spots, with no sign of suppressed numbers. The probable reason is mundane: the absence of agriculture, forestry, and hunting.

Smaller organisms fared worse in some studies. One, in March 2007, found fewer nesting birds, and fewer species, where radiation was more intense. Another, in March 2009, found pollinators and spiders less plentiful where exposure was greater. A third, in March 2014, found leaves decomposing about 40 per cent less at the most polluted sites, impeding the return of nutrients to the soil.

Counting heads cannot guarantee the health of individual animals, and how much damage low but continual doses inflict remains contentious, as studies clash over methods. The burgeoning herds may trumpet the dividend of human departure, but the sluggish rot of fallen leaves keeps a quieter ledger of radiation's cost.

中文译文

切尔诺贝利是灾难的代名词，却栖息着数量惊人的野生动物。1986年4月26日凌晨，苏联乌克兰境内这里的一座反应堆爆炸，放射性尘降物散布到欧洲各地。4月27日，附近普里皮亚季的约4.5万居民在数小时内撤离。半径约30公里的区域被列为禁区，白俄罗斯也将相邻地带围封起来。

2015年10月，《当代生物学》发表的一项研究结合白俄罗斯一侧的空中调查与足迹计数。当地驼鹿、狍、马鹿和野猪的密度，与四个未受污染保护区的同类相当；狼的数量则为对照保护区的七倍多。较早的直升机计数显示，灾后十年里，驼鹿、狍和野猪数量激增。

后来的实地研究为这些发现提供了支持。2016年4月，自动感应相机在危险程度最高的一些地点附近拍到狼、野猪和貉，没有发现数量受抑制的迹象。可能的原因其实很普通：农业、林业和狩猎活动消失了。

但一些研究发现，较小的生物境况更差。2007年3月的一项研究发现，辐射较强的地方，筑巢鸟类的数量和鸟种都较少。2009年3月的另一项研究发现，辐射暴露更高的地方，传粉动物和蜘蛛较少。2014年3月的第三项研究发现，污染最严重地点的落叶分解量少了约40%，妨碍养分回到土壤。

清点动物数量，不能保证每一个体都健康。持续低剂量辐射究竟造成多大伤害，仍有争议，各项研究在方法上也存在分歧。不断扩大的兽群，或许彰显了人类离开带来的好处；而落叶迟缓的腐烂，则默默记下了辐射的另一笔代价。

核心词汇

先记这22项。连着下方的原文短语一起回想。

synonymous with 形容词短语
是……的代名词
is synonymous with disaster

disperse v.
散布；扩散
dispersing radioactive fallout

radioactive adj.
放射性的
radioactive fallout across Europe

fallout n.
放射性尘降物
radioactive fallout across Europe

inhabitant n.
居民
about 45,000 inhabitants

comparable adj.
相当的；可比较的
comparable in density

density n.
密度
comparable in density

contamination n.
污染
free of contamination

fieldwork n.
实地研究
Later fieldwork bolstered

bolster v.
支持；加强证据
fieldwork bolstered these findings

hazardous adj.
危险的
the most hazardous spots

absence n.
不存在；缺少
the absence of agriculture

organism n.
生物
Smaller organisms fared worse

exposure n.
辐射暴露
where exposure was greater

decompose v.
分解；腐烂
leaves decomposing about 40 per cent

impede v.
阻碍
impeding the return of nutrients

guarantee v.
保证
guarantee the health

continual adj.
持续的
low but continual doses

inflict v.
造成伤害
low but continual doses inflict

contentious adj.
有争议的
remains contentious

dividend n.
好处；红利
the dividend of human departure

sluggish adj.
缓慢的
the sluggish rot of fallen leaves

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

disaster n.
灾难

startling adj.
惊人的

wildlife n.
野生动物

reactor n.
核反应堆

be emptied of 动词短语
被撤空；人员被全部迁走

radius n.
半径

off-limits adj.
禁止进入的

fence off 动词短语
围封；用围栏隔开

adjoining adj.
相邻的

terrain n.
地带；地形

Current Biology 专名
《当代生物学》

aerial adj.
空中的

footprint n.
足迹

elk n.
驼鹿

roe deer 名词短语
狍

boar n.
野猪

to prep.
与；引出比较对象

reserve n.
自然保护区

numerous adj.
数量众多的

helicopter n.
直升机

elk n.
驼鹿

surge v.
迅速增加

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

post-disaster adj.
灾后的

snap v.
拍下照片

suppress v.
抑制；压低

probable adj.
很可能的

mundane adj.
普通的；平常的

agriculture n.
农业

forestry n.
林业

nest v.
筑巢

intense adj.
强烈的

pollinator n.
传粉动物

plentiful adj.
数量多的；丰富的

polluted adj.
受污染的

individual adj.
个体的

how much damage 名词短语
多大伤害

clash over 动词短语
在……问题上存在分歧

burgeoning adj.
迅速增长的

herd n.
兽群

trumpet v.
大力宣扬；彰显

departure n.
离开

rot n.
腐烂

ledger n.
账本；此处比喻记录

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Chernobyl is _____ disaster, yet it holds a startling amount of wildlife.

02 Later fieldwork _____ these findings.

03 The probable reason is mundane: the _____ of agriculture, forestry, and hunting.

04 Counting heads cannot _____ the health of individual animals

文章主线

- 灾后 1986年切尔诺贝利核事故后，居民撤离、周边封禁，受污染地区却出现大量野生动物。
- 繁盛 白俄罗斯一侧多种兽类密度与四个未污染保护区相当，狼的数量为对照区的七倍多。
- 原因 可能的解释是农耕、伐木和狩猎消失，人类活动减少给野生动物带来了生存空间。
- 代价 部分研究发现高辐射处筑巢鸟数量及鸟种较少，传粉昆虫和蜘蛛也少；最重污染处落叶少分解约40%，阻碍养分回土。
- 限度 动物多不等于个体健康；长期低剂量辐射造成多大伤害仍有争议，不能据此说辐射无害。

参考答案　1 synonymous with 2 bolstered 3 absence 4 guarantee